

Lecture No (4)

Internet Frameworks

Objectives:

- To encapsulate what developers have learned over the past twenty years while programming sites and applications for the web.
- To reveal insights into recent trends and technologies being used to build large scale and high-performance web applications while keeping them as easy-to-maintain.
- To understand web developer to know robust web applications for PC, as well as mobile devices.



Outcomes:

- Web developer will understand basic concepts of web frameworks being used in building quality, robust and dynamic web applications
- Web developer will understand which web framework to choose that fulfills all your requirements to develop an application at least ten times faster with new web framework/technology than a typical earliest framework.



Outline

1. Introduction
2. Why web frameworks?
3. Functionality
4. Architecture of Web Framework
5. Types of Web frameworks
6. Most popular Frameworks for Web Applications
7. What are the takeaways?
8. Question and Answer



Web frameworks referred as 'Web application frameworks' or 'a software framework'

Web framework is tool or a piece of software that offers a way to create and run web applications

Is created to support the development of dynamic sites, web services and web applications and publishing of web applications and web sites

A web framework is a code library that makes web development quicker and easier by giving basic patterns for building reliable, scalable and maintainable web applications.

Designed to support the development of web applications including web services, web resources and web APIs



ProBytes

WHAT IS A
WEB FRAMEWORK?





To help streamline and speed up the process, and successful frameworks



To act as a support system for the development and deployment of applications



It reduces time and energy in developing web applications through sanitization process, session and error handling and authentication logic



To allow designers and developers to focus on building an unique feature for their web app

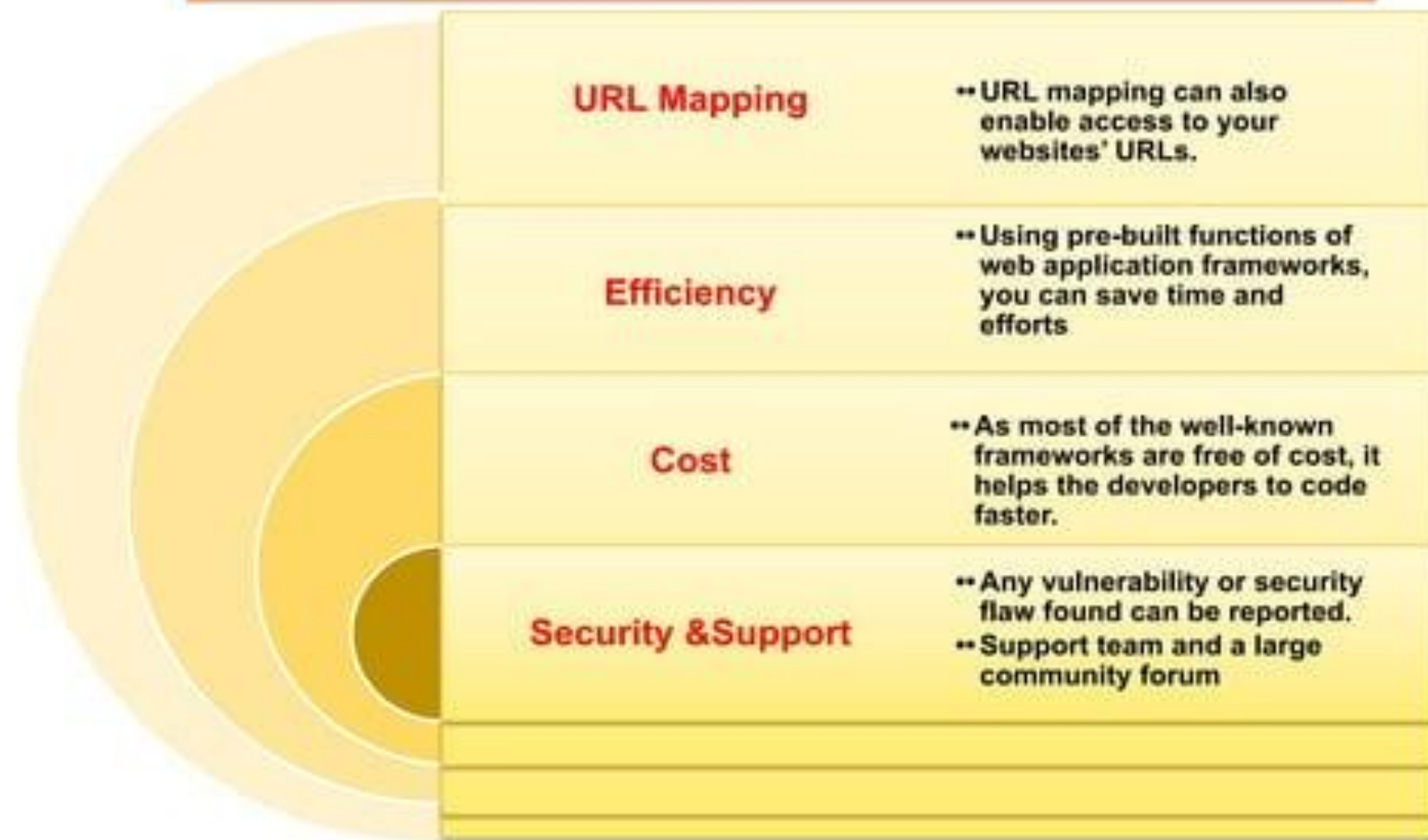


To provide templates and session management and database access libraries.

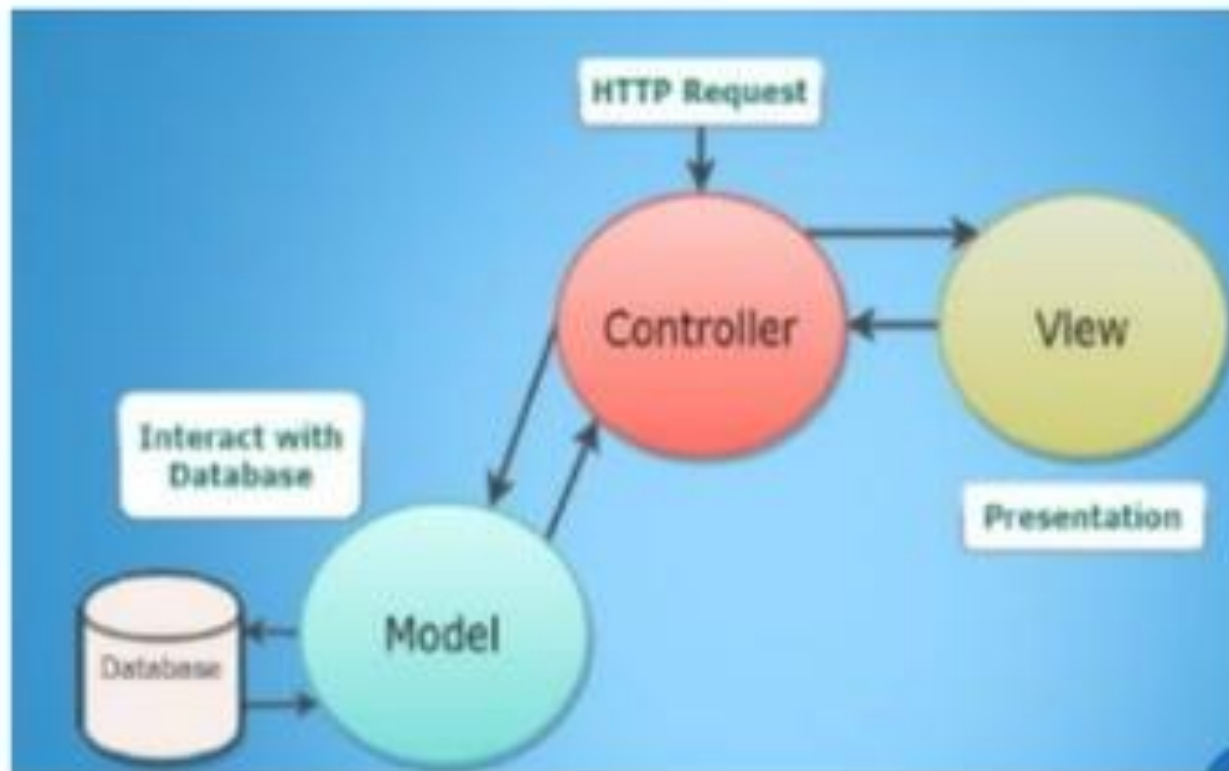
Why are Web Application Frameworks useful?

Web Framework

Functionalities:

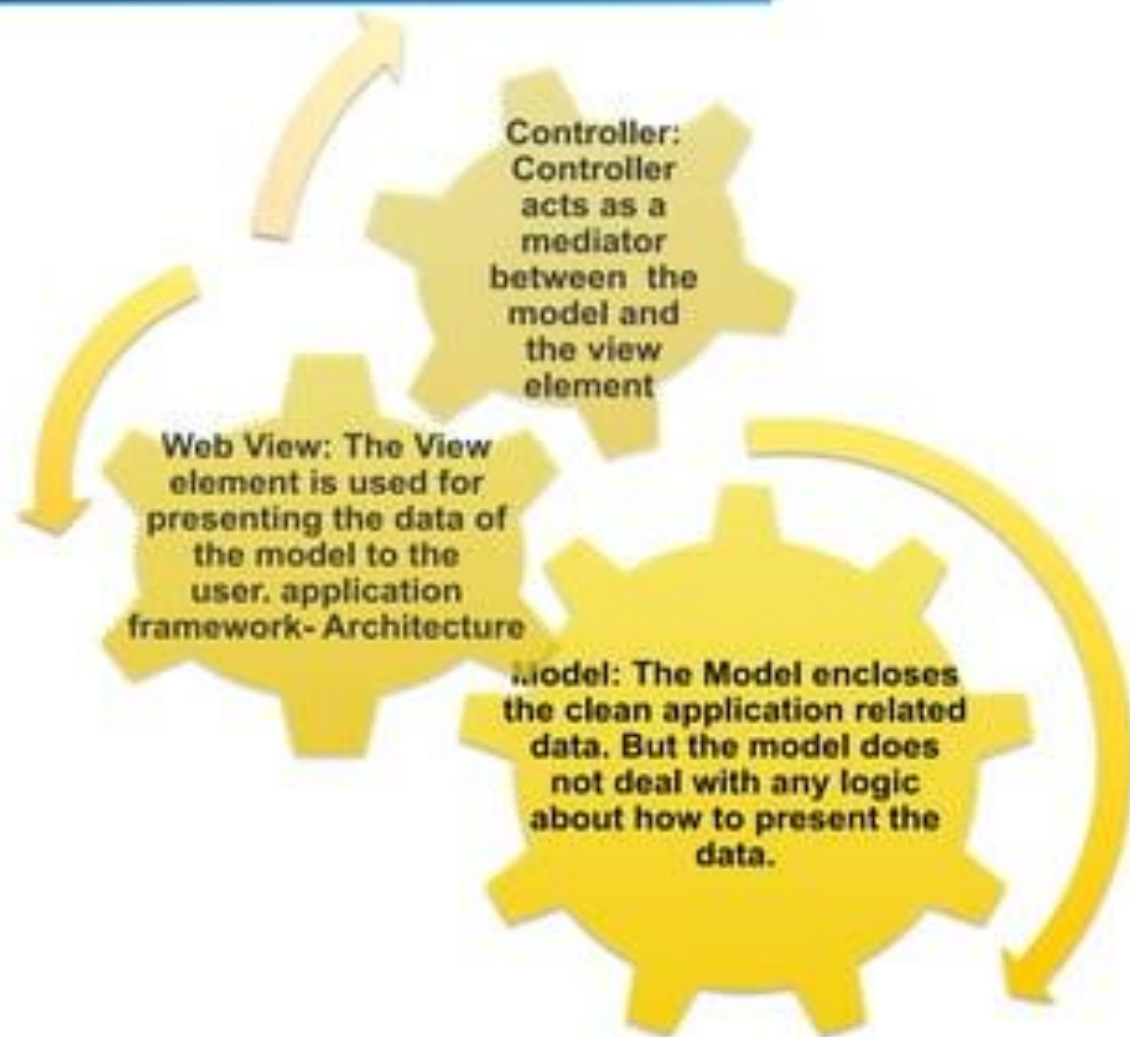


Web application framework- Architecture



- **Model View Controller** is a design pattern created for developing applications specifically web applications
- Creation of application takes different aspects
 - ☐ UI Logic
 - ☐ Input logic
 - ☐ Business Logic

What is MVC?



Benefits of MVC Architecture:

- ☐ Logical clustering of related acts on any controller can be achieved through MVC
- ☐ Developers work concurrently on different parts the same application-controller, model and the views part
- ☐ Models can have numerous views.

Types of Web frameworks

Server-side Frameworks

Client-side Frameworks

Back-End Frameworks

Django (Python)
NodeJs(Javascript)
Express (JS, NodeJS)
Ruby on Rails (Ruby)

Server-side frameworks handle HTTP requests, database control and management, URL mapping.

Front-End frameworks

Angular,
Ember.JS,
React.JS,
Vue.JS,

Function inside the browser

A number of animated features can be created with frontend and single page applications.

Web Development Frameworks in 2020

Frontend frameworks



React JS



Angular JS



Vue JS



Ember JS



BACKBONE.JS

Backend frameworks



django



Flask



Laravel



Ruby on Rails



About React JS

- Enhanced Speed & Performance
- Unidirectional Data Binding
- Flexibility across Web & Mobile
- Provides Virtual DOM
- Ease in Learning & Expanding Unidirectional Data Binding
- Highly compatible with vast Libraries
- Stable Code

Popular Web applications using React JS framework :-





'Node's goal is to provide an easy way to build scalable network programs'

Node.js is **'server-side JavaScript'**

- ❑ Core in c++ on top of Google's v8 engine
- ❑ Node.js is JavaScript running on the server-side
- ❑ It's NOT a web framework, and it's also NOT a language
- ❑ **Open Source**
- ❑ Programs for Node.js are written in JavaScript but not in the same JavaScript we are use to.
- ❑ Everything inside Node.js runs in a **single-thread**
- ❑ It makes use of **event-loops** via JavaScript's **callback** functionality to implement the non- blocking I/O

Popular Web applications using Node JS framework :-



Popular Web applications using React JS framework :-

Upwork

IBM

The Guardian

PayPal

Forbes



freelancer



Angular JS

- Offers great flexibility
 - Two-way Data binding
 - Modular development structure
 - Ease in Maintaining
 - Dependency Injection
 - TypeScript-based
 - High performance for Single Page Hefty Apps



Features Node Js

- Fast processing with Robust technology stack
- Seamless JSON Support
- MVP development
- Simple & Easy to learn
- Event-based model with rich ecosystem
- Effortless scalability



- Standalone web server for development & testing
- Stimulates the custom web application development
- Batteries-included web framework
- MVC (Model-View-Controller) Paradigm
- Compatible with vast OS and Databases
- Enhanced Security
- Huge proactive community

Popular Web applications using django framework :-



Take Aways /Conclusion:

Web application framework simplifies and speeds-up the development process significantly, helping you create a highly responsive and engaging web app.

In recent technology there are multiple tools that make web application development easier and depending upon your need you may choose one web framework that fulfills all your requirements or embed multiple frameworks.

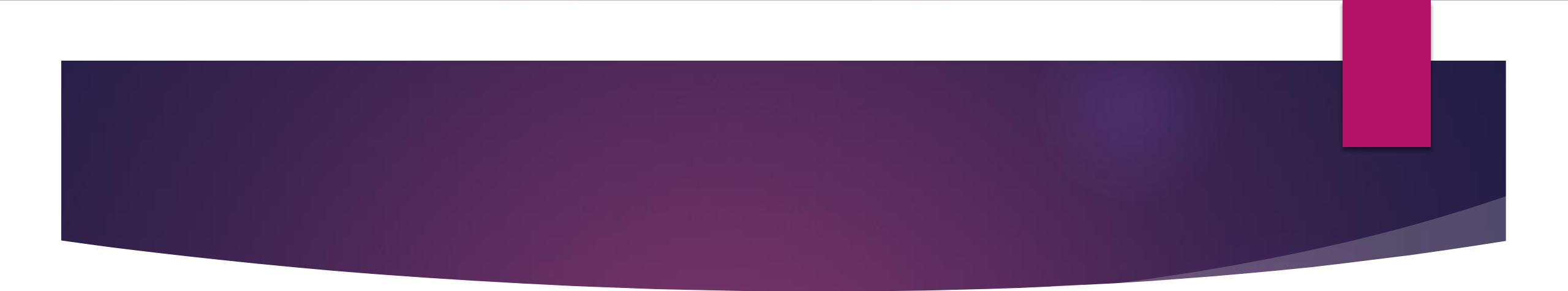
In backend, Python has begun to show more dominance in recent times, gaining control and beating PHP with the eliteness of Django and Flask. But Laravel has still vital significance and will stay at least for the coming few years. While Ruby on Rails has also made its place in the list of best backend web development frameworks.

Along with Machine learning and AI deliverables, Python can be seen growing exponentially while JavaScript & PHP remained consistent over the period of time.

JavaScript is tyrannizing the frontend web development and showing no signs of going down in near future.

With their immense capabilities at advancing web and mobile-apps development, Node.js frameworks make the developing apps effortless.

With continually evolving technologies, scrutinizing our project requirement and resource availability is key in choosing the right framework that delivers the best results.



The End